Time-Varying Rician K-factor in Measured Vehicular Channels at cmWave and mmWave Bands

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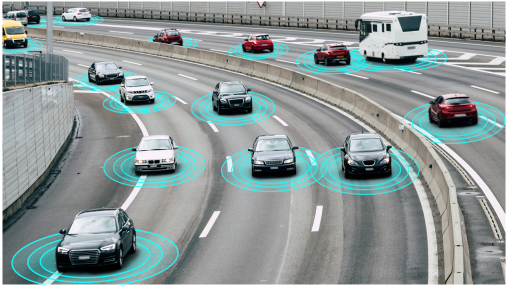
Technische Universität Wien

Institute of Telecommunications

Introduction

- increasing demand for high data rate transmission



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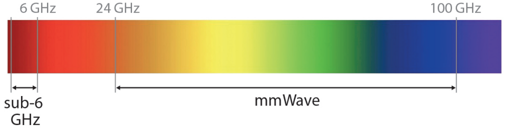
- mmWave bands allow for significantly higher data rates



Motivation

- comparative multi-band measurements are necessary
- multi-band propagation well investigated in vehicular scenarios
  - path loss, blockage loss, angular spread, RMS delay and Doppler spread

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  - crucial for designing transmission and reception techniques

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- Ricean K-factor
  - one key parameter to characterize small-scale fading
  - crucial for designing transmission and reception techniques
  - how the K-factor varies across these frequency bands ?

Contribution

- **comparative analysis of the time-varying Rician K-factor**
  - cmWave and mmWave frequency bands

[2] M. Hofer et al. "Wireless vehicular multiband measurements in centimeterwave and millimeterwave bands" IEEE 32nd International Symposium on Personal, Indoor and Mobile Radio Communications (PIMRC 2021)

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- **comparative analysis of the time-varying Rician K-factor**
  - cmWave and mmWave frequency bands
  - multi-band channel measurements in an urban environment [1]

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Contribution

- **comparative analysis of the time-varying Rician K-factor**
  - cmWave and mmWave frequency bands
  - multi-band channel measurements in an urban environment [1]
- **investigate the correlation between the K-factor and the RMS delay spread.**
  - compare the estimated correlation coefficient with the one defined in 3GPP

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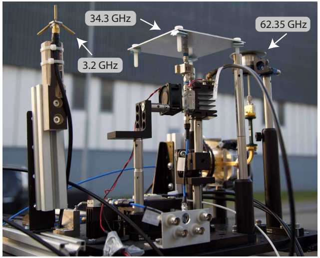
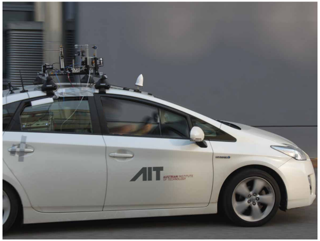
  - custom-built omnidirectional monopole
  - mounted on a car rooftop





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- measurement setup
  - 3.2 GHz, 34.3 GHz and 62.35 GHz
  - measurement bandwidth of 155.5 MHz
- transmit antennas
  - custom-built omnidirectional monopole
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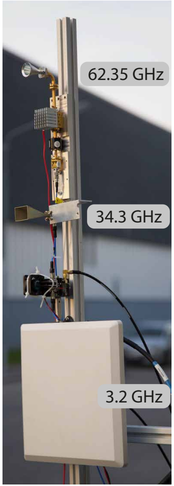
- receive antennas
  - directional antennas with 18° HPBW
  - mounted on a tripod



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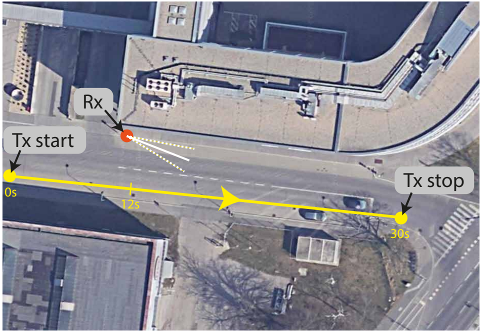
Measurement Campaign

- wireless channel measurements
  - moving vehicle with Tx antennas
  - static Rx antennas mounted on a tripod
  - moving vehicle stops at the intersection



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- measurement procedure
  - channel sounding using OFDM
  - measurement parameters
  - sequence divided into snapshots
  - first OFDM symbol used as cyclic prefix

Parameter	**Value**
Carrier Frequency f_c	3.2, 34.3, 62.35 GHz
Number of Subcarriers K	311
Subcarrier Spacing $\triangle f$	500 kHz
Bandwidth B	155.5 MHz
Symbol Duration t_s	2 µs
Symbols per Snapshot N_sym	5
Interval Between Snapshots t_R	31.25 µs
Snapshot Duration t_snap	10 µs
Number of Snapshots N	960 000
Measurement Duration t_m	30 s

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- result: time-variant channel transfer function

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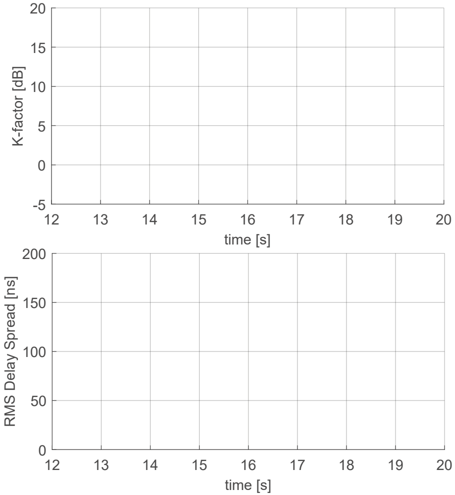

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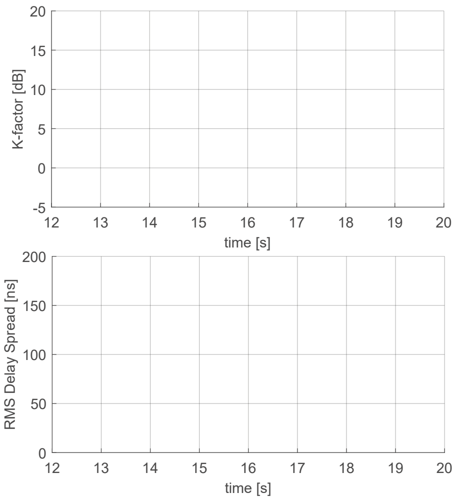

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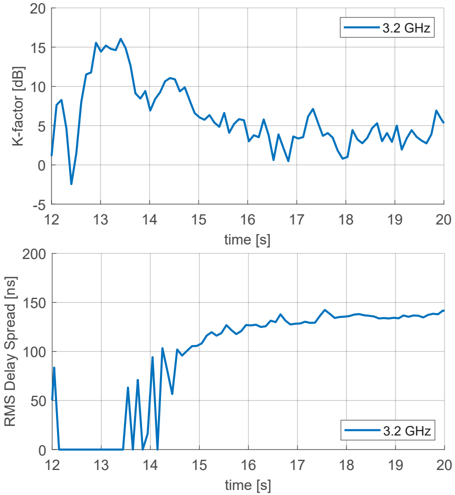

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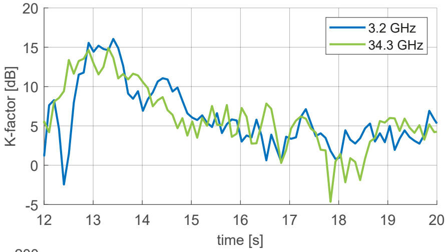

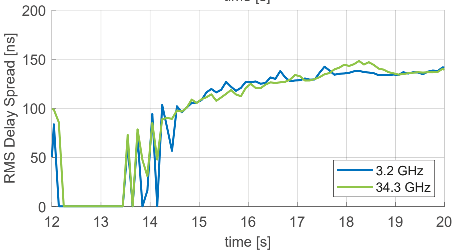

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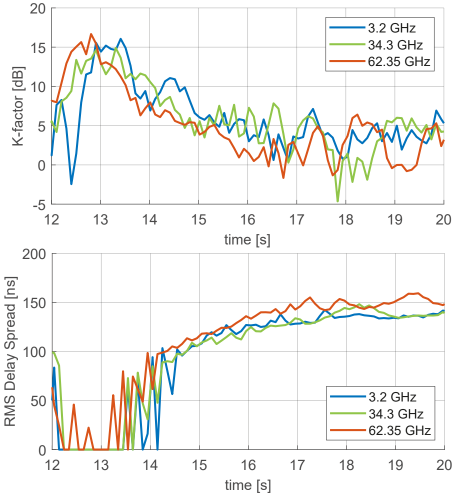

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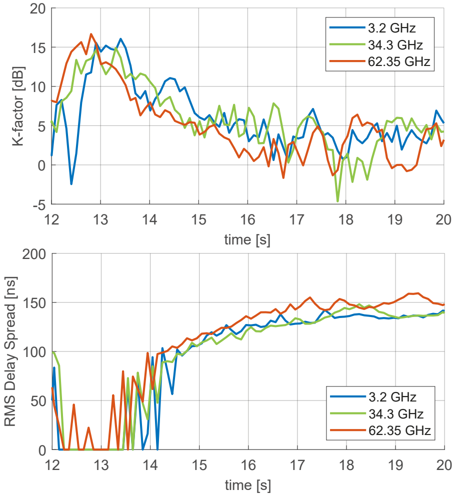

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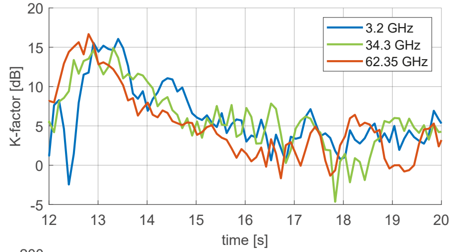

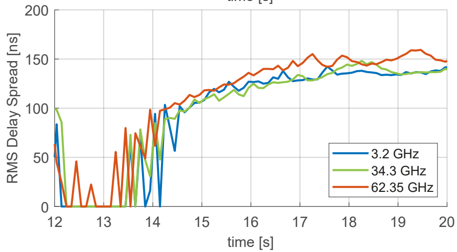

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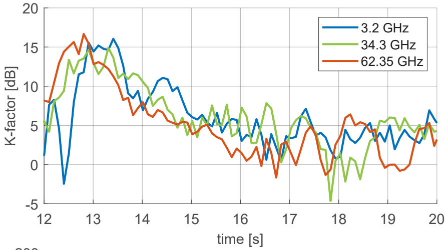
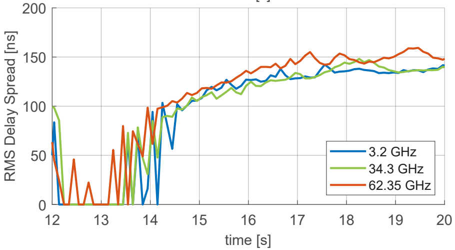
  - 12 to 13 s: Tx car passes Rx
  - 13 to 20 s: Tx car moves away from Rx
  - clear relationship between K-factor and RMS delay spread





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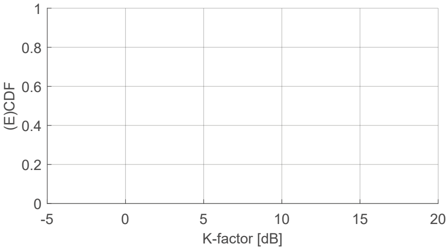

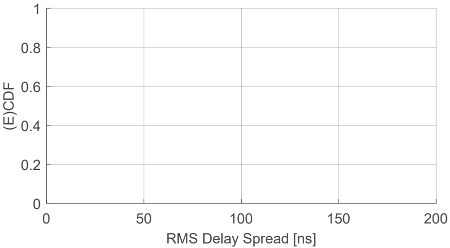

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	Estimated					3GPP
	Mean	Std.	Mean	Std.		
3.2 GHz	4.93	3.98	123.66	49.87	-0.498	-0.7
34.3 GHz	5.49	4.25	119.63	49.05	-0.703	-0.7
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σ_τ . . . RMS delay spread
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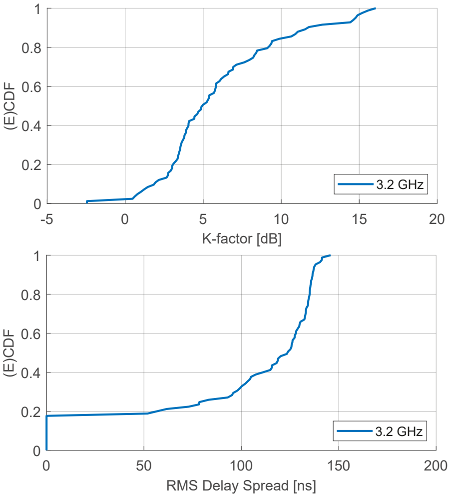

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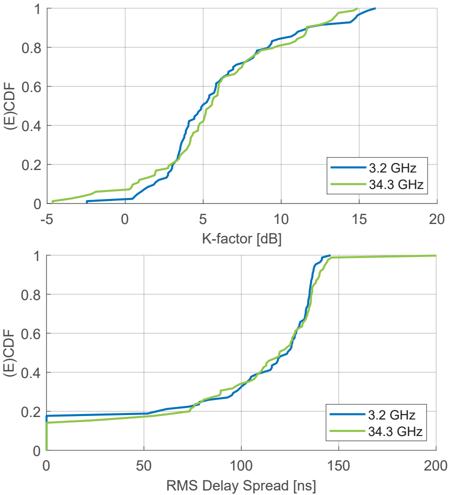

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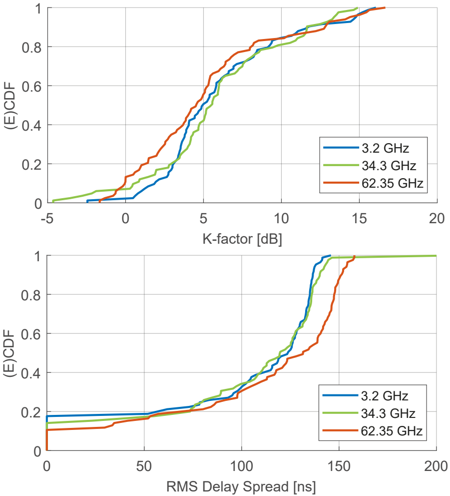

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Summary and Conclusion

- minor differences in the K-factor and RMS delay spread across different frequency bands
  - **similar propagation conditions at both cmWave and mmWave frequencies**
- K-factor and RMS delay spread are inversely related
  - **as the K-factor decreases, the RMS delay spread increases and vice versa**
  - **frequency-dependent correlation coefficient**

Thank You !

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K-factor Estimation #1

- K-factor estimated using method of moments technique
- first calculate the power of the time-variant channel

$$P_{\mathrm{H}}^{(i)}[k,n] = \left|\mathrm{H}^{(i)}[k,n]\right|^2 \tag{1}$$

- first moment, or average power of the time-variant channel is given as

$$\overline{P}_{\mathrm{H}}^{(i)} = \frac{1}{KN_{\mathrm{stat}}} \sum_{k=0}^{K-1} \sum_{n=0}^{N_{\mathrm{stat}}-1} P_{\mathrm{H}}^{(i)}[k,n] \tag{2}$$

- second moment of interest is the RMS fluctuation of $P_{\mathrm{H}}^{(i)}[k,n]$ about $\overline{P}_{\mathrm{H}}^{(i)}$ is given by

$$\upsilon_{\mathrm{P}}^{(i)} = \sqrt{\frac{1}{KN_{\mathrm{stat}}} \sum_{k=0}^{K-1} \sum_{n=0}^{N_{\mathrm{stat}}-1} \left(P_{\mathrm{H}}^{(i)}[k,n] - \overline{P}_{\mathrm{H}}^{(i)}\right)^2} \tag{3}$$

K-factor Estimation #2

- next, the power of the constant channel term is computed as

$$\left|V^{(i)}\right|^2 = \sqrt{\left(\overline{P}_{\mathrm{H}}^{(i)}\right)^2 - \left(\upsilon_{\mathrm{P}}^{(i)}\right)^2} \tag{4}$$

- the power of the fluctuating channel term is given by

$$\left(\sigma^{(i)}\right)^2 = \overline{P}_{\mathrm{H}}^{(i)} - \left|V^{(i)}\right|^2 \tag{5}$$

- finally, the estimated K-factor for the i-th stationarity region is given by the ratio of the constant to the fluctuating channel term, expressed as

$$K^{(i)} = \frac{\left|V^{(i)}\right|^2}{\left(\sigma^{(i)}\right)^2} \tag{6}$$

Estimating Local Scattering Function

- using the time-variant CTF $\mathrm{H}^{(i)}[k, n]$, we first estimate the local scattering function (LSF) given as

$$\mathcal{C}^{(i)}[\tau, \nu] = \frac{1}{IJ} \sum_{w=0}^{IJ-1} \left| \mathcal{H}_w^{(i)}[\tau, \nu] \right|^2 \tag{7}$$

with the Doppler index $\nu \in \{-N_{\mathrm{stat}}/2, \ldots, N_{\mathrm{stat}}/2 - 1\}$ and the delay index $\tau \in \{0, \ldots, K-1\}$

- the windowed frequency response is

$$\mathcal{H}_w^{(i)}[\tau, \nu] = \sum_{k=-K/2}^{K/2} \sum_{n=-N_{\mathrm{stat}}/2}^{N_{\mathrm{stat}}/2-1} \mathrm{H}[k, n + iN_{\mathrm{stat}}] \cdot \mathrm{G}_w[k, n] \, \mathrm{e}^{-\mathrm{j}2\pi(\nu n - \tau k)}, \tag{8}$$

where the tapers $\mathrm{G}_w[k, n]$ are two-dimensional discrete prolate spheroidal sequences

[2] L. Bernado et al., "Delay and Doppler spreads of nonstationary vehicular channels for safety-relevant scenarios", IEEE Transactions on Vehicular Technology, vol. 63, no. 1, 2014.

RMS Delay Spread Estimation

- calculate the power delay profile (PDP) as the expectation of the LSF over the Doppler domain, given by

$$\mathcal{P}^{(i)}[\tau] = \frac{1}{N_{\mathrm{stat}}} \sum_{\nu=-N_{\mathrm{stat}}/2}^{N_{\mathrm{stat}}/2-1} \mathcal{C}^{(i)}[\tau,\nu] \tag{9}$$

- we calculate the RMS delay spread $\sigma_\tau^{(i)}$ as second-order moment of $\mathcal{P}^{(i)}[\tau]$ given by

$$\sigma_\tau^{(i)} = \sqrt{\frac{\sum_{\forall\tau} \tau^2 \mathcal{P}^{(i)}[\tau]}{\sum_{\forall\tau} \mathcal{P}^{(i)}[\tau]} - \left(\frac{\sum_{\forall\tau} \tau \mathcal{P}^{(i)}[\tau]}{\sum_{\forall\tau} \mathcal{P}^{(i)}[\tau]}\right)^2} \tag{10}$$

Correlation Coefficient Estimation

- compute the correlation coefficient between the K-factor $K^{(i)}$ and the RMS delay spread $\sigma_\tau^{(i)}$ given by

$$\rho = \frac{\sum_{i=0}^{L_{\mathrm{stat}}-1}\left(K^{(i)}-\overline{K}\right)\left(\sigma_\tau^{(i)}-\overline{\sigma_\tau}\right)}{\sqrt{\sum_{i=0}^{L_{\mathrm{stat}}-1}\left(K^{(i)}-\overline{K}\right)^2}\sqrt{\sum_{i=0}^{L_{\mathrm{stat}}-1}\left(\sigma_\tau^{(i)}-\overline{\sigma_\tau}\right)^2}}, \tag{11}$$

where $\overline{K}=\frac{1}{L_{\mathrm{stat}}}\sum_{i=0}^{L_{\mathrm{stat}}-1}K^{(i)}$ and $\overline{\sigma_\tau}=\frac{1}{L_{\mathrm{stat}}}\sum_{i=0}^{L_{\mathrm{stat}}-1}\sigma_\tau^{(i)}$ represent the mean values of the K-factor and the RMS delay spread, respectively